The gist of it is shown in figure 15.18: it doesn't involve bounding boxes anymore, rather we want to find the intersection point of the lines passing through the segments, and then check that it lies within both segments. There are, actually, three possible cases:

1. The intersection of the two lines, point `P`, lie outside of both segments;
2. `P` lies inside one of the two segments, but not both;
3. `P` lies within both segments.

The third case is the only one where we have an intersection.

How do we find `P`? Well, first, more precisely, we will use vectors and semi-lines rather than just lines.

We define vectors $v=BA=(B_x-A_x, B_y-A_y)$ and $w=DC=(D_x-C_x, D_y-C_y)$: these vectors start from the second point in the segment and end at the first one; now, remember that vectors can be translated seamlessly, so, for instance, we can move the v so that it starts where `w` ends, or vice versa, and compute their sum or product. For instance, considering the vector `u` that is shown in figure 15.19.A (and that we'll define in a few lines), we show it sharing point `A` with v out of convenience, so that it's apparent how the two vectors are orthogonal.